dense plasma to a cooling gas of protons and electrons that can finally come together to make neutral atoms, allowing light to travel freely between them instead of constantly bouncing off the charged particles. We call the end of this fireball stage of the early universe the “surface of last scattering,” because it’s a kind of surface *in time* at which light goes from being trapped in plasma to traveling long distances across the cosmos.

This is what we see when we look at the cosmic microwave background: the moment that delineates the end of the Hot Big Bang, and the transition to a universe in which space is dark and silent and light travels through it. It is also the beginning of the cosmic Dark Ages—the time when the gas is slowly cooling and condensing into clumps, drawn in by the tiny density blips set up by those initial fluctuations. Sometime around the hundred-million-year mark, one of those clumps becomes so dense it is able to ignite into a star, and Cosmic Dawn has officially begun.

COSMIC DAWN

The transition from a dark, gaseous universe to one shimmering with the light of galaxies and stars was driven, largely, by a kind of matter so exotic that we haven’t been able to re-create it even in the most powerful particle colliders. In the mix with the radiation, hydrogen gas, and a sprinkling of other primordial elements was a substance we know today as *dark matter*. It’s not really dark, but rather invisible: seemingly unwilling to interact with light in any way. No emission of radiation, no absorption, no reflection. A light beam heading toward a clump of dark matter, as far as we can tell, passes right through. But where dark matter really shines[fn17] is in its ability to gravitate. When regular matter tries to condense in a clump drawn in by its own gravity, that matter has pressure, and pushes back. But dark matter can condense without feeling this force. A side effect of not interacting with light is not interacting much with anything at all, since, under most circumstances, collisions between particles of matter come from electrostatic repulsion, which needs interaction with light to take place. (Photons are particles of light, but they’re also the carriers of electromagnetic force, so if something is invisible, it doesn’t experience electromagnetic attraction or repulsion.) No electromagnetism, no pressure.

The first little blips of higher-density matter, set down by the fluctuations at the end of inflation, contained a mix of radiation, dark matter, and